ANALOG NEURAL NETWORK TEST BOARD (70x100mm)

DIY Photolithography & Photoresist PCB Etching Guide - ISO A5 Format

1. BOARD SPECIFICATIONS & PRINT RULES

* Dimensions: 70.0 mm height x 100.0 mm width (4:3 aspect ratio)
* Substrate: Standard 1.6 mm FR-4 (1 oz / 35 um Copper Clad)
* Page Format: ISO A5 (148 x 210 mm) - Landscape
* CRITICAL: Print at 100% SCALE (Actual Size). DO NOT FIT TO PAGE!
* Caliper Check: Verify 50.0 mm scale bar with digital calipers.

2. PHOTORESIST PAGE DIRECTORY

Page 2: Front Copper (F.Cu) - POSITIVE RESIST (Normal View)
Page 3: Front Copper (F.Cu) - POSITIVE RESIST (Mirrored - Emulsion Down)
Page 4: Front Copper (F.Cu) - NEGATIVE DRY FILM (Normal View)
Page 5: Front Copper (F.Cu) - NEGATIVE DRY FILM (Mirrored View)
Page 6: Component Silkscreen, Drilling Map & Reference Legend

3. CHEMICAL PROCESSES & RECIPES

A. Positive Pre-Sensitized Boards (Bungard / Positiv 20):

* Artwork: Solid black traces block UV (Pages 2 & 3).
* Developer: 10 g / L Sodium Hydroxide (NaOH) at 20 C (60-90 s).

B. Negative Dry Film Photoresist (Riston Blue Film):

* Artwork: Transparent traces transmit UV to cure (Pages 4 & 5).
* Developer: 10 g / L Sodium Carbonate (Na2CO3) (90-120 s).

C. Etching Baths:

* Ferric Chloride (FeCl3) at 45-50 C with continuous aeration.
* Alternatively: 2 parts Hydrogen Peroxide (3%) + 1 part Muriatic Acid.

4. DRILLING & VIAS GUIDE

*** Vias & Jumpers: 0.80 mm carbide bit.**
*** Header Pins (J_IN, J_PWR, J_OUT): 1.00 mm bit.**
*** Mounting Holes (H1..H4): 3.20 mm bit (M3 hardware).**

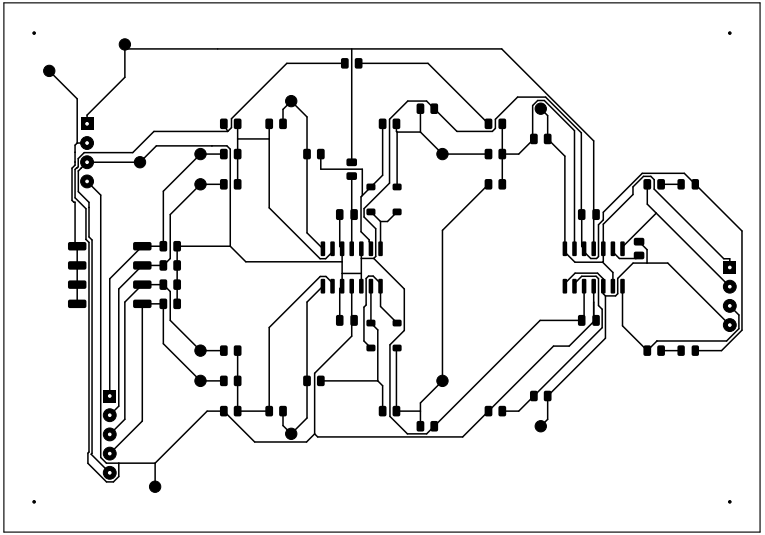
PRO PHOTOLITHOGRAPHY TIP: EMULSION-TO-COPPER CONTACT

Always print the Mirrored View (Page 3 for Positive, Page 5 for Negative) onto transparency film so that the toner/ink emulsion directly contacts the photoresist surface during UV exposure. This eliminates light undercutting beneath traces, yielding pin-sharp 0.25 mm tracks and clean pads. Check transparency density with a strong light - double up two printed sheets if necessary.

Use standard 365-405 nm UV LED exposure box for 45-120 seconds depending on lamp wattage.

ANALOG NEURAL NETWORK TEST BOARD (70x100mm) — PHOTORESIST MASK

FORMAT: ISO A5 (210x148 mm) | SCALE: 1:1 TRUE SIZE | 100% SINGLE COPPER LAYER (0-OHM JUMPERS)



50.0 mm [1:1 CALIPER CALIBRATION SCALE]

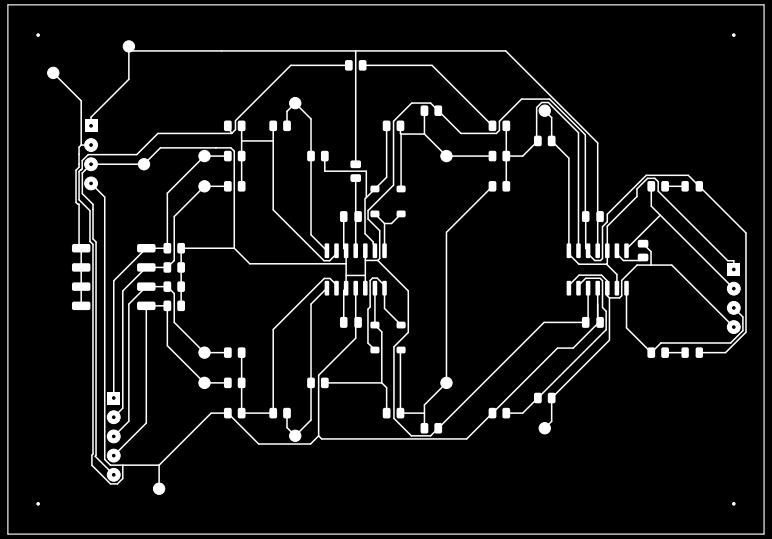
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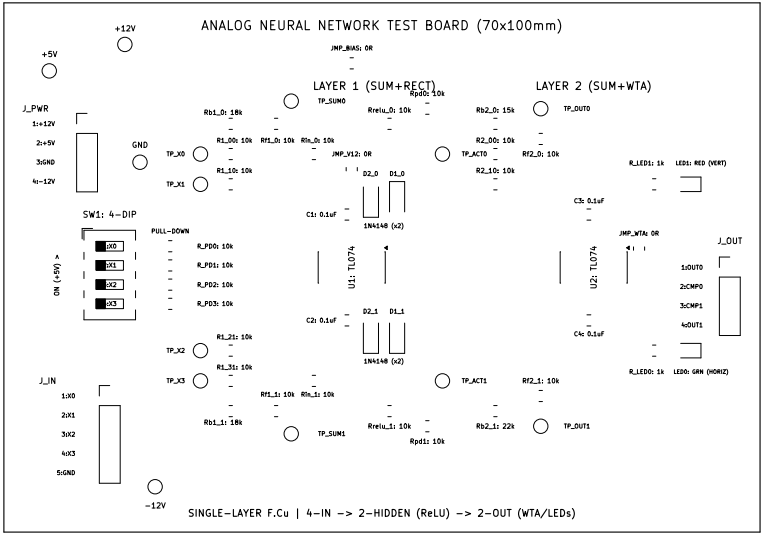


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ANALOG NEURAL NETWORK TEST BOARD (70x100mm)
+12V
+5V
GND
-12V
J_PWR
1:+12V
2:+5V
3:GND
4:-12V
J_IN
1:X0
2:X1
3:X2
4:X3
5:GND
SW1: 4-DIP
ON (+5V) >
X0
X1
X2
X3
PULL-DOWN
R_PD0: 10k
R_PD1: 10k
R_PD2: 10k
R_PD3: 10k
LAYER 1 (SUM+RECT)
JMP_BIAS: OR
TP_SUM0
TP_SUM1
Rb1_0: 18k
Rb1_1: 18k
R1_00: 10k
R1_10: 10k
R1_21: 10k
R1_31: 10k
Rf1_0: 10k
Rf1_1: 10k
Rin_0: 10k
Rin_1: 10k
Rrelu_0: 10k
Rrelu_1: 10k
Rpd0: 10k
Rpd1: 10k
TP_X0
TP_X1
TP_X2
TP_X3
JMP_V12: OR
D2_0
D1_0
D2_1
D1_1
1N4148 (x2)
1N4148 (x2)
C1: 0.1uF
C2: 0.1uF
U1: TL074
LAYER 2 (SUM+WTA)
TP_OUT0
TP_OUT1
Rb2_0: 15k
Rb2_1: 22k
R2_00: 10k
R2_10: 10k
Rf2_0: 10k
Rf2_1: 10k
TP_ACT0
TP_ACT1
C3: 0.1uF
C4: 0.1uF
U2: TL074
JMP_WTA: OR
R_LED1: 1k
R_LED0: 1k
LED1: RED (VERT)
LED0: GRN (HORIZ)
J_OUT
1:OUT0
2:CMP0
3:CMP1
4:OUT1
SINGLE-LAYER F.Cu | 4-IN -> 2-HIDDEN (ReLU) -> 2-OUT (WTA/LEDs)